

Reasoning about smePMP

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Preface

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Abstract

This `abstract` environment provides a summary of the work, which may or may not be extensive. The intention is that this should be limited to 1 page.

Acronyms

CSR Control Status Register. 3

PMP Physical Memory Protection. 3

Chapter 1

Introduction

this is a test

Chapter 2

Background

2.1 Physical Memory Protection

Physical Memory Protection (PMP) is a RISC-V extension that Machine mode uses to enforce fault containment and isolation by restricting physical memory access available to lower privilege levels (User- and Supervisor-mode). These restrictions are applied in a few, up to 64, **PMP** entries. Each of these entries get two **Control Status Registers (CSRs)**, one to define the address range ($pmpaddr_i$) and one to define the permissions and addressing mode ($pmpcfg_i$). The $pmpcfg$ looks like this:

7	6	5	4	3	2	1	0
L	0	A	X	W	R		

TABLE 2.1: $pmpcfg$ layout

[1]

Bibliography

- [1] *3.1. Machine-Level ISA, Version 1.13 :: RISC-V Ratified Specifications Library*. [Online]. Available: <https://docs.riscv.org/reference/isa/priv/machine.html#pmp> (visited on 03/19/2026).

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Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

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